Storage elevates phenolic content and antioxidant activity but suppresses antiproliferative and pro-apoptotic properties of colored-flesh potatoes ...

Colored-flesh potatoes are an excellent source of health-benefiting dietary polyphenols, but are stored for up to 3-6 months before consumption. This study investigated the effect of simulated commercial storage conditions on antioxidant activity (DPPH, ABTS), phenolic content (FCR) and composition (UPLC-MS), and anticancer properties (early, HCT-116 and advanced stage, HT-29 human colon cancer cell lines) of potato bioactive compounds. Extracts from seven potato clones of differing flesh colors (white, yellow, and purple) before and after 90 days of storage were used in this study. The antioxidant activity of all clones increased with storage; however, an increase in total phenolic content was observed only in purple-fleshed clones. Advanced purple-fleshed selection CO97227-2P/PW had greater levels of total phenolics, monomeric anthocyanins, antioxidant activity and a diverse anthocyanin composition as compared with Purple Majesty. Purple-fleshed potatoes were more potent in suppressing proliferation and elevating apoptosis of colon cancer cells compared with white- and yellow-fleshed potatoes. The extracts from both fresh and stored potatoes (10-30 μg/mL) suppressed cancer cell proliferation and elevated apoptosis compared with the solvent control, but these anticancer effects were more pronounced with the fresh potatoes. Storage duration had a strong positive correlation with antioxidant activity and percentage of viable cancer cells and a negative correlation with apoptosis induction. These results suggest that although the antioxidant activity and phenolic content of potatoes were increased with storage, the antiproliferative and pro-apoptotic activities were suppressed. Thus, in the assessment of the effects of farm to fork operations on the health-benefiting properties of plant foods, it is critical to use quantitative analytical techniques in conjunction with in vitro and/or in vivo biological assays.